

Mycology & Virology

Original MCQ Reviewer with Answer Key and Rationales

Concept coverage follows the uploaded Mycology and Virology material. The question stems, scenarios, distractors, and rationales are newly written for study use rather than copied or lightly paraphrased.

8.9 Mycology - Specimens, Yeasts, and Basic Identification

- 1. A laboratory receives several specimens for fungal culture. Which specimen type is generally least desirable because it may contain little material and is easily contaminated by surrounding flora?**
 - A. Tissue biopsy
 - B. Cerebrospinal fluid
 - C. Needle aspirate
 - D. Surface swab

- 2. A technologist prepares a direct examination of skin scrapings for suspected fungal infection. Which reagent is most appropriate for clearing keratin and background debris?**
 - A. 10% KOH
 - B. India ink
 - C. Acid-fast stain
 - D. Gram iodine

- 3. An encapsulated yeast is suspected in the CSF of an immunocompromised patient. Which direct preparation can demonstrate a clear capsule surrounding the yeast?**
 - A. India ink preparation
 - B. KOH mount only
 - C. Modified acid-fast stain
 - D. Lactophenol cotton blue culture mount

- 4. A patient has a superficial infection involving hair, skin, and nails. Which group of fungi should receive strong consideration?**
 - A. Dermatophytes
 - B. Mucorales only
 - C. Dimorphic systemic fungi only
 - D. Cryptococcus species

- 5. A yeast isolate is being evaluated for presumptive identification as *Candida albicans*. Which rapid test is especially useful?**
 - A. Germ tube test
 - B. Oxidase test
 - C. Coagulase test
 - D. Acid-fast stain

6. A filament extending from a yeast cell shows a constriction where it emerges. How should this finding be interpreted?

- A. A true germ tube
- B. A pseudohyphal structure
- C. An arthroconidium
- D. A sporangium

7. Why is cornmeal agar with Tween 80 useful when evaluating an unknown yeast?

- A. It enhances demonstration of characteristic microscopic structures
- B. It detects viral antigens
- C. It measures antifungal concentration
- D. It selects only dimorphic molds

8. A CSF yeast isolate produces blastoconidia but no pseudohyphae or arthroconidia. Which pair of tests would be particularly useful if *Cryptococcus neoformans* is suspected?

- A. Urease and niger seed agar
- B. Coagulase and catalase
- C. Optochin and bile solubility
- D. Germ tube and oxidase

9. What causes the dark pigmentation of *Cryptococcus neoformans* colonies on birdseed agar?

- A. Melanin production associated with phenol oxidase activity
- B. Hemolysin production
- C. Urea degradation alone
- D. Capsule dissolution

10. Which yeast is classically germ tube positive and can form terminal chlamydoconidia?

- A. *Candida albicans*
- B. *Candida glabrata*
- C. *Rhodotorula* species
- D. *Cryptococcus neoformans*

11. A yeast forms only blastoconidia on cornmeal Tween 80 agar, without pseudohyphae or arthroconidia. Which genus best matches this pattern?

- A. *Cryptococcus*
- B. *Trichosporon*
- C. *Geotrichum*
- D. *Candida tropicalis*

12. A germ tube-negative yeast recovered from an immunocompromised patient produces pink to coral colonies and is urease positive. Which organism is most likely?

- A. *Rhodotorula* species
- B. *Candida albicans*
- C. *Geotrichum candidum*
- D. *Candida tropicalis*

13. Which yeast is capable of producing ascospores under appropriate culture conditions?

- A. *Saccharomyces cerevisiae*
- B. *Candida albicans*
- C. *Cryptococcus neoformans*
- D. *Candida glabrata*

8.9 Mycology - Dimorphic, Dermatophytic, and Opportunistic Fungi

14. A thermally dimorphic fungus is cultured at room temperature and at body temperature. In infected tissue, which form is generally expected for the dimorphic fungi described in the source?

- A. Yeast phase
- B. Mold phase
- C. Sporangial phase
- D. No fungal structures

15. Which mold produces thick-walled, barrel-shaped arthroconidia separated by empty cells in its environmental phase?

- A. *Coccidioides immitis*
- B. *Histoplasma capsulatum*
- C. *Sporothrix schenckii*
- D. *Blastomyces dermatitidis*

16. A tissue isolate of a dimorphic fungus contains elongated, cigar-shaped yeast forms. Which organism is most characteristic?

- A. *Sporothrix schenckii*
- B. *Coccidioides immitis*
- C. *Histoplasma capsulatum*
- D. *Aspergillus fumigatus*

17. Which mold phase produces large tuberculate macroconidia and may resemble *Sepedonium* microscopically?

- A. *Histoplasma capsulatum*
- B. *Candida albicans*
- C. *Cryptococcus neoformans*
- D. *Malassezia furfur*

18. A large parent yeast cell has numerous smaller buds arranged around its circumference in a 'mariner's wheel' appearance. Which organism is suggested?

- A. *Paracoccidioides brasiliensis*
- B. *Sporothrix schenckii*
- C. *Histoplasma capsulatum*
- D. *Coccidioides immitis*

19. Septate hyphae are clearly seen in a mold culture. Which major fungal group can be excluded based on the source's description?

- A. Mucorales
- B. Dermatophytes
- C. Dematiaceous molds
- D. Dimorphic molds

20. A KOH preparation of skin scales shows short hyphae mixed with round or oval budding yeast, producing a 'spaghetti and meatballs' appearance. What is the likely organism?

- A. *Malassezia furfur*
- B. *Cryptococcus neoformans*
- C. *Candida glabrata*
- D. *Histoplasma capsulatum*

21. Which dermatophyte genus may infect skin, hair, and nails?

- A. *Trichophyton*
- B. *Microsporum*
- C. *Epidermophyton*
- D. *Malassezia*

22. A dermatophyte develops a distinctive cherry-red pigment and teardrop-shaped microconidia along the hyphae. Which species is most likely?

- A. *Trichophyton rubrum*
- B. *Trichophyton tonsurans*
- C. *Microsporum audouinii*
- D. *Epidermophyton floccosum*

23. A dermatophyte produces large smooth club-shaped macroconidia but no microconidia. Which genus is most consistent?

- A. Epidermophyton
- B. Trichophyton
- C. Microsporum
- D. Aspergillus

24. A mold colony is dark brown to black on both the surface and reverse, and microscopy shows pigmented septate hyphae. How should the mold be classified?

- A. Dematiaceous mold
- B. Hyaline mold
- C. Dermatophyte only
- D. Mucorales

25. A rapidly growing mold begins white and later develops a black pepper-like surface from abundant black conidia. Which organism best fits?

- A. Aspergillus niger
- B. Aspergillus fumigatus
- C. Penicillium species
- D. Fusarium species

26. Which Aspergillus species is most commonly associated with pulmonary aspergillosis and produces phialides mainly on the upper portion of the vesicle?

- A. Aspergillus fumigatus
- B. Aspergillus niger
- C. Aspergillus terreus
- D. Aspergillus flavus

27. A hyaline mold produces rose-lavender colonies and canoe- or sickle-shaped multiseptate macroconidia. Which genus is most likely?

- A. Fusarium
- B. Phialophora
- C. Exophiala
- D. Rhizopus

28. A rapidly growing cottony nonseptate mold has rhizoids that arise between the sporangiophores rather than directly opposite them. Which organism is favored?

- A. Lichtheimia (formerly Absidia)
- B. Rhizopus
- C. Aspergillus
- D. Penicillium

29. An HIV-positive patient has encapsulated yeast in CSF, a positive urease test, and brown growth on niger seed agar. What is the most likely cause of meningitis?

- A. Cryptococcus neoformans
- B. Candida albicans
- C. Candida tropicalis
- D. Rhodotorula species

30. An immunocompromised patient's bone marrow contains small intracellular yeast-like cells. Culture at room temperature produces tuberculate macroconidia. Which fungus is most likely?

- A. Histoplasma capsulatum
- B. Coccidioides immitis
- C. Sporothrix schenckii
- D. Aspergillus fumigatus

31. A lung specimen from an immunocompromised patient shows cup-shaped cysts with GMS stain, and the organism does not grow on routine fungal media. Which organism should be suspected?

- A. Pneumocystis jirovecii
- B. Histoplasma capsulatum
- C. Candida albicans
- D. Scopulariopsis species

32. Sputum contains large spherules filled with endospores. Culture produces septate hyphae with barrel-shaped arthroconidia. Which fungus is indicated?

- A. Coccidioides immitis
- B. Histoplasma capsulatum
- C. Talaromyces marneffeii
- D. Cryptococcus neoformans

33. Which mold is identified in the source as the most common cause of human mucormycosis?

- A. Rhizopus species
- B. Candida albicans
- C. Penicillium species
- D. Scopulariopsis species

8.10 Virology - Principles and Major Viral Diseases

34. Which viral property serves as a fundamental basis for initial classification?

- A. Whether the genome contains RNA or DNA
- B. Gram reaction
- C. Ability to form spores
- D. Catalase activity

35. Which virus is a major cause of bronchiolitis and other serious respiratory disease in infants?

- A. Respiratory syncytial virus
- B. Rotavirus
- C. Norovirus
- D. Epstein-Barr virus

36. A patient develops myocarditis and pleuritic chest pain from an enteroviral infection. Which virus is strongly associated with this syndrome?

- A. Coxsackie B virus
- B. RSV
- C. CMV
- D. Rotavirus

37. Which herpesvirus can produce a mononucleosis-like illness similar to Epstein-Barr virus infection?

- A. Cytomegalovirus
- B. HSV-2
- C. Varicella-zoster virus
- D. Human papillomavirus

38. Which pair is emphasized in the source as important causes of viral pneumonia in adults?

- A. Influenza virus and adenovirus
- B. Rotavirus and norovirus
- C. HBV and HCV
- D. Coxsackie A and B only

39. A young child has severe gastroenteritis, vomiting, and dehydration. Which Reoviridae member should be considered?

- A. Rotavirus
- B. Norovirus
- C. Hantavirus
- D. Parvovirus B19

40. A child develops fever and a characteristic slapped-cheek rash. Which virus is the likely cause?

- A. Parvovirus B19
- B. Adenovirus
- C. Rotavirus
- D. Coxsackie B virus

41. Which virus is commonly responsible for outbreaks of gastroenteritis among adults after contaminated food or water exposure?

- A. Norovirus
- B. RSV
- C. Parvovirus B19
- D. CMV

42. A patient has genital warts and cervical cells showing koilocytosis. Which viral group is most closely associated?

- A. Human papillomavirus
- B. Rotavirus
- C. Rabies virus
- D. RSV

43. Which approaches may assist in diagnosing viral infections in addition to conventional viral culture?

- A. Serology, nucleic acid testing, and histologic/cytologic methods
- B. Only Gram staining
- C. Only bacterial culture
- D. Only coagulation testing

44. A patient has a reactive HIV screening test followed by a negative differentiation assay, but nucleic acid testing detects viral RNA. What does the molecular result support?

- A. HIV infection can be present before a full antibody response develops
- B. The molecular result proves the patient is uninfected
- C. Maternal IgG caused the RNA result
- D. The virus cannot be detected during early infection

45. A patient with suspected rabies encephalitis undergoes antemortem testing. Which specimen and method are described as useful for direct antigen detection?

- A. Nuchal skin biopsy with immunofluorescence
- B. Urine culture
- C. Stool antigen test
- D. Bone marrow KOH preparation

46. Which molecular method is increasingly used for rapid confirmation of influenza and other respiratory viral infections?

- A. RT-PCR
- B. Coagulase testing
- C. KOH microscopy
- D. Urease testing

8.10 Virology - Clinical Identification and Problem Solving

47. A patient has painful genital vesicles. Which specimen provides useful material for rapid direct detection of herpes simplex viral antigen?

- A. Vesicle or lesion material
- B. Stool
- C. Bone marrow
- D. Hair shaft

48. A college student has marked fatigue, pharyngitis, and lymphadenopathy, but a rapid heterophile test is negative. If other common causes have been excluded, which virus should receive additional specific testing?

- A. Epstein-Barr virus
- B. Rotavirus
- C. Rabies virus
- D. Hantavirus

49. An older gardener develops encephalitis after heavy mosquito exposure in an area where dead birds have been observed. Which virus best fits the epidemiology?

- A. West Nile virus
- B. Norovirus
- C. Parvovirus B19
- D. Rotavirus

50. A worker develops severe pulmonary illness after entering a rodent-infested building and inhaling contaminated material. Which virus should be suspected?

- A. Hantavirus
- B. Rotavirus
- C. West Nile virus
- D. EBV

51. A three-year-old develops vomiting, diarrhea, and severe dehydration during the winter. Bacterial stool cultures are negative. Which virus is especially likely?

- A. Rotavirus
- B. CMV
- C. EBV
- D. Hepatitis D virus

52. Which measurements are useful in the source's discussion of HIV treatment assessment?

- A. Viral load, CD4 lymphocyte quantitation, and resistance testing
- B. PT and APTT only
- C. Serum sodium only
- D. Fungal culture only

53. An infant with congenital heart disease develops apnea and bronchiolitis. Bacterial testing is unrevealing. Which virus is especially important to test for?

- A. RSV
- B. Norovirus
- C. Rotavirus
- D. EBV

54. A person is bitten by a suspicious wild fox that cannot be captured for testing. According to the source, what immediate management is indicated?

- A. Rabies immune globulin plus rabies vaccination
- B. Wait for neurologic symptoms
- C. Perform weekly throat cultures
- D. Administer antifungal therapy

55. Which test is emphasized for laboratory confirmation of severe acute respiratory syndrome caused by the SARS coronavirus?

- A. RT-PCR
- B. Latex agglutination
- C. Blood culture
- D. KOH preparation

56. A pregnant patient has active genital herpes lesions, and her premature infant develops a fatal disseminated infection despite negative bacterial studies. Which virus is the likely cause?

- A. HSV
- B. RSV
- C. CMV only
- D. EBV

57. An adult who had chickenpox as a child develops a localized painful vesicular eruption years later. Which virus has reactivated?

- A. Varicella-zoster virus
- B. HSV-1
- C. EBV
- D. Norovirus

58. A cruise ship experiences an outbreak of acute vomiting and diarrhea involving many adult passengers. Which virus is the most likely cause?

- A. Norovirus
- B. Rotavirus
- C. RSV
- D. Parainfluenza virus

59. An immunocompromised transplant patient develops hemorrhagic cystitis. Which virus-specimen combination is most appropriate to investigate?

- A. BK virus in urine
- B. HPV in serum
- C. EBV in stool
- D. HBV in skin

60. Which virus belongs to the filovirus group and can appear filamentous, curved, or looped on electron microscopy?

- A. Ebola virus
- B. West Nile virus
- C. Poliovirus
- D. Zika virus

61. Which mosquito-borne flavivirus is associated with congenital microcephaly and other adverse pregnancy outcomes?

- A. Zika virus
- B. West Nile virus
- C. Hepatitis C virus
- D. Norovirus

62. A pregnant traveler develops hepatitis-like illness after visiting an endemic region and tests negative for hepatitis A. Which fecal-oral hepatitis virus is particularly dangerous during pregnancy?

- A. Hepatitis E virus
- B. Hepatitis B virus
- C. Hepatitis C virus
- D. Hepatitis D virus

Answer Key & Rationales

Check this section after answering the questions.

1. D

Rationale: Swabs are generally poor specimens for fungal recovery because they may dry out, contain insufficient material, and collect contaminating skin flora.

2. A

Rationale: KOH preparations help dissolve keratinous material and other debris in skin, hair, and nail specimens, making fungal elements easier to visualize.

3. A

Rationale: India ink excludes the polysaccharide capsule of *Cryptococcus neoformans*, producing a clear halo around the yeast cell.

4. A

Rationale: Dermatophytes are specialized for keratinized tissues and commonly cause infections of skin, hair, and/or nails.

5. A

Rationale: Most *C. albicans* isolates produce true germ tubes after short incubation under appropriate conditions, making this a useful presumptive identification test.

6. B

Rationale: True germ tubes lack a constriction at their point of origin. A constriction favors pseudohyphal development and can cause false interpretation of a germ tube test.

7. A

Rationale: Cornmeal Tween 80 agar promotes formation and visualization of structures such as blastoconidia, pseudohyphae, true hyphae, arthroconidia, and chlamydoconidia.

8. A

Rationale: *C. neoformans* is urease positive and produces brown to black pigmentation on niger seed (birdseed) agar because of phenol oxidase activity.

9. A

Rationale: Phenol oxidase enables *C. neoformans* to oxidize diphenolic compounds and produce melanin, yielding brown to black colonies.

10. A

Rationale: *C. albicans* commonly produces germ tubes and characteristic thick-walled terminal chlamydoconidia under appropriate culture conditions.

11. A

Rationale: *Cryptococcus* species characteristically produce blastoconidia but lack pseudohyphae and arthroconidia.

12. A

Rationale: Rhodotorula species are notable for pink to coral pigmentation and may act as opportunistic pathogens in immunosuppressed patients.

13. A

Rationale: *S. cerevisiae* is an ascomycetous yeast and can form ascospores on appropriate sporulation media.

14. A

Rationale: The source describes dimorphic molds as occurring in the yeast form in infected tissues and at approximately 37 C, with mold growth at lower culture temperatures.

15. A

Rationale: The mold phase of *C. immitis* forms distinctive barrel-shaped arthroconidia separated by empty or ghost cells.

16. A

Rationale: *S. schenckii* produces characteristic oval to elongated cigar-shaped yeast forms in tissue.

17. A

Rationale: The mold phase of *H. capsulatum* produces large, thick-walled tuberculate macroconidia that can resemble *Sepedonium*.

18. A

Rationale: The multiple budding yeast form of *P. brasiliensis* creates the classic mariner's wheel appearance.

19. A

Rationale: Common Mucorales such as *Rhizopus* and *Mucor* display broad aseptate or sparsely septate hyphae, unlike the septate groups listed.

20. A

Rationale: The combination of short hyphae and budding yeast in skin scales is characteristic of *M. furfur*, the cause of tinea versicolor.

21. A

Rationale: Trichophyton species can involve all three keratinized sites. Microsporum typically affects skin and hair, while Epidermophyton affects skin and nails.

22. A

Rationale: *T. rubrum* is distinguished by its characteristic red pigmentation and compatible microscopic morphology.

23. A

Rationale: Epidermophyton species characteristically lack microconidia and form club-shaped macroconidia.

24. A

Rationale: Dematiaceous molds contain dark pigment in their hyphae and typically produce dark colonies.

25. A

Rationale: *A. niger* produces black conidia that create a pepper-like colony surface while the reverse remains relatively pale.

26. A

Rationale: *A. fumigatus* is a major cause of aspergillosis and typically sporulates from the upper half to two-thirds of the vesicle.

27. A

Rationale: *Fusarium* species are hyaline molds recognized by their characteristic curved, multiseptate macroconidia.

28. A

Rationale: The source distinguishes *Lichtheimia* from *Rhizopus* by the location of the rhizoids relative to the sporangiophores.

29. A

Rationale: Encapsulation, urease positivity, and melanin-producing brown colonies on niger seed agar strongly support *C. neoformans*.

30. A

Rationale: *H. capsulatum* is thermally dimorphic, appears as small intracellular yeast in tissue, and produces tuberculate macroconidia in its mold phase.

31. A

Rationale: *P. jirovecii* is identified in respiratory material using special stains such as GMS and is not recovered on routine fungal culture media.

32. A

Rationale: *C. immitis* forms spherules containing endospores in tissue and highly infectious barrel-shaped arthroconidia in its mold phase.

33. A

Rationale: *Rhizopus* species are highlighted as the most common cause of mucormycosis, particularly in patients with serious underlying disease.

34. A

Rationale: Viruses are initially classified in part by their nucleic acid composition, with viral genomes containing either RNA or DNA.

35. A

Rationale: RSV is a major pediatric respiratory pathogen and is especially important in infants, causing bronchiolitis, pneumonia, and related syndromes.

36. A

Rationale: Coxsackie B viruses are classically associated with myocarditis, pericarditis, and pleurodynia.

37. A

Rationale: CMV may cause a self-limited mononucleosis-like syndrome, especially in otherwise healthy individuals.

38. A

Rationale: Influenza viruses and adenoviruses are important causes of adult respiratory disease, including pneumonia.

39. A

Rationale: Rotavirus is an important cause of gastroenteritis in infants and young children and can produce significant dehydration.

40. A

Rationale: Parvovirus B19 causes erythema infectiosum, also known as fifth disease, which classically produces a slapped-cheek rash.

41. A

Rationale: Noroviruses cause community and institutional outbreaks of acute gastroenteritis across age groups, including adults.

42. A

Rationale: HPV causes anogenital warts, and infected squamous cells may show koilocytosis. Certain HPV types are associated with anogenital neoplasia.

43. A

Rationale: The source describes serology, nucleic acid methods such as PCR/probes, and cytologic or histologic examination as useful alternatives to culture.

44. A

Rationale: Nucleic acid methods can detect viral RNA during early infection before antibodies become sufficiently detectable.

45. A

Rationale: Rabies antigen can be demonstrated by direct immunofluorescence in cutaneous nerves around hair follicles from a nuchal skin biopsy.

46. A

Rationale: RT-PCR provides sensitive and rapid detection of respiratory viruses and has increasingly replaced slower culture-based approaches.

47. A

Rationale: Material collected directly from an active vesicle can be examined using rapid antigen methods, and molecular methods can also detect HSV sensitively.

48. A

Rationale: EBV-specific testing may help establish infectious mononucleosis when a rapid heterophile assay is negative, especially when the clinical syndrome remains suggestive.

49. A

Rationale: Birds are important reservoirs for West Nile virus and mosquitoes transmit infection to incidental human hosts, in whom neurologic disease may occur.

50. A

Rationale: Hantavirus pulmonary syndrome is associated with exposure to aerosolized rodent excreta and can progress to severe respiratory disease and pulmonary edema.

51. A

Rationale: Rotavirus commonly causes significant winter gastroenteritis in infants and young children.

52. A

Rationale: The source discusses plasma viral load, CD4 lymphocyte counts, and genotype/phenotype resistance testing as relevant laboratory information for antiviral management.

53. A

Rationale: RSV is a major cause of bronchiolitis and hospitalization for respiratory illness in infants, with severe disease risk in vulnerable children.

54. A

Rationale: When a potentially rabid animal cannot be tested, the source recommends prompt postexposure prophylaxis with rabies immune globulin and vaccine.

55. A

Rationale: The source identifies RT-PCR as the recommended laboratory method for confirming SARS coronavirus infection.

56. A

Rationale: The source describes neonatal disseminated HSV, particularly HSV-2 associated with maternal genital infection, as potentially fatal.

57. A

Rationale: Varicella-zoster virus remains latent in sensory ganglia after chickenpox and may later reactivate as shingles.

58. A

Rationale: Norovirus is a classic cause of explosive gastroenteritis outbreaks in settings such as cruise ships and affects adults as well as children.

59. A

Rationale: BK polyomavirus has urinary tract tropism and can cause renal and bladder disease in immunocompromised patients; urine is an appropriate specimen.

60. A

Rationale: Ebola is a filovirus with characteristic filamentous forms and is associated with severe viral hemorrhagic fever.

61. A

Rationale: The source links Zika virus infection with serious fetal neurologic abnormalities, including microcephaly.

62. A

Rationale: Hepatitis E is transmitted by the fecal-oral route and is associated with a substantially increased risk of severe or fatal disease during pregnancy.